Introduction to the ThreatMetrix Platform

ThreatMetrix enables visibility into user activities across all digital interactions and devices. Our platform provides historic context that helps detect user accounts being misused in real time. ThreatMetrix builds a comprehensive digital identity to authenticate each connecting user by combining information relating to devices, anonymized identity information, behavior patterns and threat intelligence. Transactions can be verified in real time against trusted patterns of behavior: high-risk anomalies are accurately identified for review while genuine users experience minimal friction. The ThreatMetrix Digital Identity Network provides the intelligence that fuels the ThreatMetrix platform. The Network harnesses global shared intelligence from millions of daily consumer interactions including logins, payments and new account originations. Using this information, ThreatMetrix stitches together a user's true digital identity by analyzing the myriad connections between devices, locations and anonymized personal information. Transactions are verified in real time against trusted patterns of behavior: high-risk anomalies are accurately identified for review while genuine users experience minimal friction.

Verticals

Our solutions aid customers in many business verticals, including:

- E-Commerce
- Banking and Brokerage
- Payments and Lending
- Gaming
- Media
- Insurance
- Government
- Healthcare
- Crowdfunding

Solutions

The ThreatMetrix Platform supports the following solutions:

1. **Payment Fraud**: detecting the use of stolen or fraudulent payment credentials in real time.
2. **Identity Verification**: ensuring that connecting users are who they say they are. This detects fraudsters attempting to open new accounts with stolen or spoofed identities that might otherwise go undetected using static identity assessment alone.
3. **Digital Authentication**: effectively recognizing new and returning users; providing a slick and streamlined login experience. Step-up authentication procedures are only used for high-risk transactions, reducing friction for trusted users and operational costs for businesses.
4. **Account Security**: protecting trusted user accounts from fraudulent account takeover, whether from stolen credentials, malware, advanced phishing scams, remote access Trojans (RATs) or automated botnet attacks.
5. **Compliance Assurance**: providing businesses with identity and location certainty, for example ensuring that businesses are not transacting with individuals or networks from countries that are subject to penalty sanctions (OFAC).

6. **Workforce Authentication**: protect business platforms from fraudulent takeover, as well as allowing a workforce to login remotely, whether on desktop or mobile.
7. **Threat Intelligence**: combining threat detection with global threat intelligence of known fraudsters and botnet participation to help proactively detect complex attack vectors.
8. **Identity Analytics**: using behavioral analytics to inform smarter machine learning models to better predict future fraud.
9. **Mobile App Security**: protecting the integrity of host mobile applications, as well as protecting the surrounding mobile platform from device spoofing and malware attacks.

Components

The following key components comprise the ThreatMetrix Platform, and are documented in this system:

- **ThreatMetrix Web**: Integrates customer's web pages into the ThreatMetrix Platform.
- **ThreatMetrix SDK**: Integrates customer's native apps for Android, iOS, OS X and Windows into the ThreatMetrix Platform.
- **ThreatMetrix API**: A family of APIs used to connect customer's systems to the ThreatMetrix Platform.
- **ThreatMetrix Portal**: A web interface used by fraud managers, analysts and security teams to configure, manage and review their ThreatMetrix implementation.
- **ThreatMetrix Reporting**: A web interface used by customers to view detailed information and statistics from the ThreatMetrix Platform.
- **ThreatMetrix Connect**: A highly configurable integration hub that enables customers to connect their systems to third party APIs via the ThreatMetrix Platform.

For more information, please visit www.threatmetrix.com

Implementation Guides

APIs

Introduction to the ThreatMetrix® APIs

ThreatMetrix® offers several APIs that enable customers to connect their systems to the ThreatMetrix® Platform. The API links will correspond to the user's region: https://h-api-sb.online-metrix.net for the EU and https://h-api.online-metrix.net for all other regions. All ThreatMetrix API requests are authenticated using the ThreatMetrix® Dynamic Decision Platform (DDP) API Security.

Supported API Request formats in the post body:

- JSON (some API calls only work with JSON)
- url encoded

JSON example:

```
curl -X POST https://h-api.online-metrix.net/api/session-query -H "Content-Type: application/json" -d '{"org_id":"orgid123","api_key":"your1apikey12345","session_id":"c6b4116faa8a4e6da52df6b419ef0dfc","service_type":"basic","event_type":"login","account_name":"test","account_email":"test@email.com"}'
```

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Session Query API

The Session Query API is the main API in the ThreatMetrix® platform. It is used to trigger a risk assessment of a specific event performed by a user (ThreatMetrix® refers to these events as sessions).
Purpose:

- Trigger a risk assessment of a session.
- Return the outcome of the risk assessment.
- Enables customers to pass additional attributes regarding the session, such as user information, transaction/purchase information, shipping information and more.

Documentation: Session Query API

Attribute Query API

The Attribute Query API is used to obtain information and perform a risk assessment for an entity in the ThreatMetrix® platform.
Purpose:

- Obtain information on an entity
- Trigger a risk assessment of an entity. A policy can be specified to be executed during the attribute query.

Documentation: Attribute Query API

Update API

The Update API enables customers to provide feedback regarding an event or entity, as well as manage Trust Tags and Lists.
Purpose:

- Set, Update and Remove Trust Tags on a transaction directly via the API via a Request ID.
- Providing retrospective Truth Data regarding the outcome of a session.
- Add and Remove attributes to the Block/Safe/Watch Lists directly via the API for faster maintenance of lists without the need to login to the portal and add values manually.
- Update the Review Status of a transaction.

Documentation: Update API

Query API

The Update API enables customers to fetch information regarding an event or entity, including Trust Tags and Lists.
Purpose:

- Check Trust Tags.
- Check if an attribute is in the Block/Safe/Watch Lists.
- Check the final review status of a transaction.

Documentation: Query API

Session Query API

The Session Query API allows customers to obtain information regarding a user's session. Once a device has been profiled, the attributes gathered during profiling are stored in the ThreatMetrix® Profiling Server Cache. The customer then must perform a HTTPS API Call, known as a Session Query, to retrieve information regarding the session.

The main objective of the API is to trigger the ThreatMetrix® Rules Engine to perform an analysis of the user session and to return detailed information that allows customers to make informed real-time decisions on whether a session should be considered legitimate or fraudulent. Furthermore, customers may pass in transaction related identifiers about the session, such the user's email address or credit card information which enables the Rules Engine to perform additional analysis using these extra points of data providing an even more thorough assessment of the session.

Information returned includes:

1. **The Review Status of the session**: The Review Status of a transaction which is dependent upon the execution of the rules in the policy and the policy score obtained. It may be **Pass**, **Reject**, **Review**, or **Challenge**.

2. **The Policy Score**: The policy score which is calculated as the sum of the individual risk weights of its rules.
3. **Reason Codes:** The reason codes which indicate the rules that triggered during policy evaluation.

> info
>
> **Note**
> Profiling information is stored in the Profiling Server Cache for 24 hours. The Session Query must be made within this time-frame, otherwise, the Session Query will return 'Unknown Session'.

Session Query URL

EU: https://h-api-sb.online-metrix.net/api/session-query
Other regions: https://h-api.online-metrix.net/api/session-query

Session Query with Financial Crime Digital Intelligence

Please see FCDI Extension to the Session Query for use of the Session Query with the Financial Crime Digital Intelligence product suite.

Performing a Session Query

The Session Query is a RESTful HTTPS API. The Session Query API is triggered by the customer's web application at the time of the transaction such as during login, registration and payment, depending on the implementation. The Session Query is implemented as HTTPS requests (GET or POST) with the input parameters being URL encoded and passed as part of the URL (HTTP GET) or body (HTTP POST). The HTTPS POST Session Query is recommended as it is more secure than the GET method. Please see the Session Query Examples section below for an example of each type of Session Query.

Passing in additional data

During the Session Query call, we recommend that you pass in as many additional transaction related identifiers as possible, such as the email address, account login, credit card information, transaction amount and the like. These transaction identifiers will further enhance the risk assessment for the transaction by leveraging the ThreatMetrix® Network which contains information gathered from all ThreatMetrix® customers.

When inputting parameters, please note that any preceding and trailing whitespaces will be stripped to ensure data consistency. Also, please note that the API does not accept <, >, or % characters. These are simply ignored and removed from the Input.

ThreatMetrix® API will validate the maximum length of the provided input attributes using the UTF-8 characters, and not in bytes. For example, previously, the limit 40 meant that only 40 bytes could be provided. Therefore a string which contained 30 two-byte or longer characters (as it happens with all non-ASCII characters) could be rejected, because even though it had less than 40 characters, it had

more than 40 bytes. With this change, this string will be accepted, because now the length measurement is done in UTF-8 characters.

See the Session Query Parameters section below for a full list of Mandatory and Recommended Input and Output parameters.

Sample Request

API: session-query
POST command
Headers:

- x-org-id
- x-api-key

```
curl -X POST https://h-api.online-metrix.net/api/session-query --header "x-org-id: ********" --header "x-api-key: *****************" --header "Content-Type: application/json; charset=utf-8" -d '{"session_id":"c6b4116faa8a4e6da52df6b0dfc","service_type":"all","event_type":"login","account_name":"john doe"}'
```

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Output Formats

The response format can be controlled either by setting the content-type or by passing the output_format which may be one of following:

- JSON
- url encoded
- text

Portal Event

Once the session query has completed, the transaction is available as an event in the portal within a few minutes. These events may be accessed at any time from the portal and are useful for further analysis, but more importantly, to add or remove entities to the local or global blocklists and safelists to enhance future risk assessment.

> info
> **Note**
> See "What are TMX's data retention practices?" in the FAQ for more information on how API responses are stored.

Session Query Examples

Example 1: HTTPS GET Basic Session Query

```
https://h-api.online-metrix.net/api/session-query?org_id=orgid123&api_key=your1apikey12345&session_id=3q94qw8ds79c283a49&service_type=session-policy&event_type=payment
```

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Example 2: HTTPS GET Passing account_login, account_email and password_hash during the Session Query:

```
https://h-api.online-metrix.net/api/session-query?org_id=orgid123&api_key=your1apikey12345&session_id=3q94qw8ds79c283a49&service_type=session-policy
&account_login=a.test&account_email=a.test@threatmetrix.com&password_hash=4241B68FBC12414a94DF141761614813&event_type=payment
```

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Example 3: HTTPS POST Basic Session Query

```
POST /api/session-query HTTP/1.0
Host: h-api.online-metrix.net
Content-Type: application/x-www-form-urlencoded
Content-Length: 23
org_id=orgid123&api_key=your1apikey12345&session_id=3q94qw8ds79c283a49&service_type=session-policy&event_type=payment
```

Session Query Output Formats

The following session query output formats are supported:

- text
- JSON

Text Example:


```
account_name=john%20doe&account_name_assert_history=NEGATIVE_HISTORY&account_name_attributes=_CHALLENGED&account_name_attributes=_CHALLENGE_FAILED&account_name_attributes=_CHALLENGE_PASSED&account_name_first_seen=2021%2d06%2d16&account_name_last_event=2023%2d05%2d11&account_name_last_update=2023%2d05%2d11&account_name_result=success&account_name_score=0&account_name_worst_score=%2d55&api_call_datetime=2023%2d05%2d11%2005%3a58%3a14%2e375&api_caller_ip=1%2e1%2e1%2e1&api_type=session%2dquery&api_version=14%2e2&bb_assessment=523%2e48&bb_assessment_rating=neutral&bb_bot_rating=low&bb_bot_score=500%2e00&bb_fraud_rating=neutral&bb_fraud_score=500%2e00&champion_request_duration=3&digital_id_result=not_enough_attribs&enabled_services=TMX100%2cTMX201&event_datetime=2023%2d05%2d11%2005%3a58%3a14%2e375&event_type=login&org_id=orgid123&osl_model_1_p_score=0%2e011&osl_model_1_policy=OSL_model_20200401%2d20201020&osl_model_2_p_score=0%2e039&osl_model_2_policy=OSL_model_20200701%2d20200719&osl_model_2_reason_code=po_dummy_rule&policy=default&policy_details_api=%7b%22policy_deta
```

il_api%22%3a%5b%7b%22type%22%3a%22champion%22%2c%22id%22%3a%220%22%2c%22smart
learning%22%3a%7b%22score%22%3a%2214%22%2c%22pvid%22%3a%221703600%22%2c%22sma
rt_learning_fraud_rating%22%3a%22very_low%22%2c%22smart_learning_p_score%22%3
a%220%2e000%22%2c%22smart_learning_risk_rank%22%3a%229%2e6307385698815%22%2c%
22smart_learning_raw_p_score%22%3a%221%2e2113728329619e%2d05%22%2c%22smart_le
arning_policy_version_id%22%3a%221703600%22%2c%22rules%22%3a%5b%7b%22rid%22%3
a%22292971839%22%2c%22reason_code%22%3a%22score_offset%22%2c%22score%22%3a%22
%2d100%22%7d%2c%7b%22rid%22%3a%22292971840%22%2c%22reason_code%22%3a%22advanc
ed_model%22%2c%22score%22%3a%22%2d100%22%7d%5d%7d%2c%22customer%22%3a%7b%22sc
ore%22%3a%220%22%2c%22pvid%22%3a%221683088%22%2c%22review_status%22%3a%22revi
ew%22%2c%22risk_rating%22%3a%22low%22%2c%22rules%22%3a%5b%7b%22rid%22%3a%2228
0784865%22%2c%22reason_code%22%3a%22SL%20Execution%20Policy%201%22%2c%22score
%22%3a%220%22%2c%22pvid%22%3a%220%22%2c%22rules%22%3a%5b%7b%22rid%22%3a%22148
894010%22%2c%22reason_code%22%3a%22score_calibration%22%2c%22score%22%3a%22%2
d100%22%7d%5d%7d%2c%7b%22rid%22%3a%22280784866%22%2c%22reason_code%22%3a%22SL
%20Execution%20Policy%202%22%2c%22score%22%3a%220%22%2c%22pvid%22%3a%220%22%2
c%22rules%22%3a%5b%7b%22rid%22%3a%22144245482%22%2c%22reason_code%22%3a%22po_
dummy_rule%22%2c%22score%22%3a%22%2d100%22%7d%2c%7b%22rid%22%3a%22144245484%2
2%2c%22reason_code%22%3a%22score_calibration%22%2c%22score%22%3a%22%2d100%22%
7d%5d%7d%2c%7b%22rid%22%3a%22280784867%22%2c%22reason_code%22%3a%22Condition%
22%2c%22score%22%3a%220%22%7d%2c%7b%22rid%22%3a%22280784868%22%2c%22reason_co
de%22%3a%22Call%20Policy%22%2c%22score%22%3a%220%22%2c%22pvid%22%3a%221181588
%22%2c%22rules%22%3a%5b%7b%22rid%22%3a%2237345230%22%2c%22reason_code%22%3a%2
2Call%20Policy%22%2c%22score%22%3a%22%2d100%22%2c%22pvid%22%3a%221181594%22%2
c%22rules%22%3a%5b%7b%22rid%22%3a%2237348196%22%2c%22reason_code%22%3a%22Alwa
ys%20True%22%2c%22score%22%3a%22%2d100%22%7d%2c%7b%22rid%22%3a%2237348248%22%
2c%22reason_code%22%3a%22Profiling%20Blocked%22%2c%22score%22%3a%22%2d10%22%7
d%2c%7b%22rid%22%3a%2237348276%22%2c%22reason_code%22%3a%22Profiling%20Incomp
lete%22%2c%22score%22%3a%220%22%7d%5d%7d%5d%7d%5d%7d%7d%5d%7d&policy_engine_v
ersion=14%2e2%2e1&policy_score=0&primary_industry=banking&reason_code=SL%20Ex
ecution%20Policy%201&reason_code=SL%20Execution%20Policy%202&reason_code=Cond
ition&reason_code=Always%20True&reason_code=Profiling%20Blocked&reason_code=P
rofiling%20Incomplete&reason_code=Call%20Policy&request_duration=8&request_id
=64cf40b6%2d9729%2d4082%2da7fb%2d3f709e7bb75e&request_result=success&review_s
tatus=review&risk_rating=low&secondary_industry=brokerage&service_type=all&se
ssion_id=madeupsessionidnotlinkedtoanyprofiling&session_id_query_count=2&smar
t_learning_fraud_rating=very_low&smart_learning_risk_rank=9%2e6307385698815&s
ummary_risk_score=0&tmx_reason_code=_Threat_Intelligence_Real_Location_Detect
ed&tmx_reason_code=_ACC_NAME_GBL_VEL_73_120_120_120&tmx_reason_code=_ACC_NAME
_GBL_AGE_GT_3MTHS&tmx_reason_code=_ACC_NAME_LCL_VEL_3_3_3_3&tmx_reason_code=_
ACC_NAME_LCL_AGE_LT_HR&tmx_reason_code=_TMX_LOCAL_ONLY_&tmx_reason_code=_EXPR
ESSION_ERROR&tmx_reason_code=_TMX_GBL_TT_LIMIT_REACHED_ACC_NAME&tmx_risk_rati
ng=neutral&unknown_session=yes

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JSON Example:


```
1 {
2    "account_name":"john doe",
3    "account_name_assert_history":[
4       "NEGATIVE_HISTORY"
5    ],
6    "account_name_attributes":[
```

```
7        "_CHALLENGED",
8        "_CHALLENGE_FAILED",
9        "_CHALLENGE_PASSED"
10    ],
11    "account_name_first_seen":"2021-06-16",
12    "account_name_last_event":"2023-05-11",
13    "account_name_last_update":"2023-05-11",
14    "account_name_result":"success",
15    "account_name_score":"0",
16    "account_name_worst_score":"-55",
17    "api_call_datetime":"2023-05-11 05:50:32.083",
18    "api_caller_ip":"**.**.**.**",
19    "api_type":"session-query",
20    "api_version":"14.2",
21    "bb_assessment":"523.48",
22    "bb_assessment_rating":"neutral",
23    "bb_bot_rating":"low",
24    "bb_bot_score":"500.00",
25    "bb_fraud_rating":"neutral",
26    "bb_fraud_score":"500.00",
27    "champion_request_duration":"5",
28    "digital_id_result":"not_enough_attribs",
29    "enabled_services":[
30       "TMX100",
31       "TMX201"
32    ],
33    "event_datetime":"2023-05-11 05:50:32.083",
34    "event_type":"login",
35    "org_id":"orgid123",
36    "osl_model_1_p_score":"0.011",
37    "osl_model_1_policy":"OSL_model_20200401-20201020",
38    "osl_model_2_p_score":"0.039",
39    "osl_model_2_policy":"OSL_model_20200701-20200719",
40    "osl_model_2_reason_code":[
41       "po_dummy_rule"
42    ],
43    "policy":"default",
44    "policy_details_api":{
45       "policy_detail_api":[
46          {
47             "type":"champion",
48             "id":"0",
49             "smartlearning":{
50                "score":"14",
51                "pvid":"1703600",
52                "smart_learning_fraud_rating":"very_low",
53                "smart_learning_p_score":"0.000",
54                "smart_learning_risk_rank":"9.6307385698815",
55                "smart_learning_raw_p_score":"1.2113728329619e-05",
56                "smart_learning_policy_version_id":"1703600",
57                "rules":[
58                   {
```

```
59                      "rid":"292971839",
60                      "reason_code":"score_offset",
61                      "score":"-100"
62                   },
63                   {
64                      "rid":"292971840",
65                      "reason_code":"advanced_model",
66                      "score":"-100"
67                   }
68                ]
69             },
70             "customer":{
71                "score":"0",
72                "pvid":"1683088",
73                "review_status":"review",
74                "risk_rating":"low",
75                "rules":[
76                   {
77                      "rid":"280784865",
78                      "reason_code":"SL Execution Policy 1",
79                      "score":"0",
80                      "pvid":"0",
81                      "rules":[
82                         {
83                            "rid":"148894010",
84                            "reason_code":"score_calibration",
85                            "score":"-100"
86                         }
87                      ]
88                   },
89                   {
90                      "rid":"280784866",
91                      "reason_code":"SL Execution Policy 2",
92                      "score":"0",
93                      "pvid":"0",
94                      "rules":[
95                         {
96                            "rid":"144245482",
97                            "reason_code":"po_dummy_rule",
98                            "score":"-100"
99                         },
100                        {
101                           "rid":"144245484",
102                           "reason_code":"score_calibration",
103                           "score":"-100"
104                        }
105                     ]
106                  },
107                  {
108                     "rid":"280784867",
109                     "reason_code":"Condition",
110                     "score":"0"
```

```
                    },
                    {
                        "rid":"280784868",
                        "reason_code":"Call Policy",
                        "score":"0",
                        "pvid":"1181588",
                        "rules":[
                            {
                                "rid":"37345230",
                                "reason_code":"Call Policy",
                                "score":"-100",
                                "pvid":"1181594",
                                "rules":[
                                    {
                                        "rid":"37348196",
                                        "reason_code":"Always True",
                                        "score":"-100"
                                    },
                                    {
                                        "rid":"37348248",
                                        "reason_code":"Profiling Blocked",
                                        "score":"-10"
                                    },
                                    {
                                        "rid":"37348276",
                                        "reason_code":"Profiling Incomplete",
                                        "score":"0"
                                    }
                                ]
                            }
                        ]
                    }
                ]
            }
        }
    ]
},
"policy_engine_version":"14.2.1",
"policy_score":"0",
"primary_industry":"banking",
"reason_code":[
    "SL Execution Policy 1",
    "SL Execution Policy 2",
    "Condition",
    "Always True",
    "Profiling Blocked",
    "Profiling Incomplete",
    "Call Policy"
],
"request_duration":"11",
"request_id":"e79e924f-a80f-4811-af4b-5ed00bd79588",
"request_result":"success",
```

```
  "review_status":"review",
  "risk_rating":"low",
  "secondary_industry":"brokerage",
  "service_type":"all",
  "session_id":"madeupsessionidnotlinkedtoanyprofiling",
  "session_id_query_count":"1",
  "smart_learning_fraud_rating":"very_low",
  "smart_learning_risk_rank":"9.6307385698815",
  "summary_risk_score":"0",
  "tmx_reason_code":[
     "_Threat_Intelligence_Real_Location_Detected",
     "_ACC_NAME_GBL_VEL_73_120_120_120",
     "_ACC_NAME_GBL_AGE_GT_3MTHS",
     "_ACC_NAME_LCL_AGE_LT_HR",
     "_TMX_LOCAL_ONLY_",
     "_EXPRESSION_ERROR",
     "_TMX_GBL_TT_LIMIT_REACHED_ACC_NAME"
  ],
  "tmx_risk_rating":"neutral",
  "unknown_session":"yes"
}
```


Session Query Response

The session query returns detailed information about the transaction based on various factors that include:

1. Information collected during the current profiling session.
2. The reason codes that triggered the rules in the policy which was run.
3. The policy score for the transaction which determines its risk rating.
4. Information about the attributes and entities generated from analyzing the profiling data, the most important of which are the device ids which are generated by ThreatMetrix® and form a core part of risk assessments.
5. The Review Status which determines the action to take such as Pass, Reject and Review.

Some Example Session Query Response are shown below:

URL Encoded Session Query Response

This is the typical format for a Session Query Response. Each returned parameter is separated by a '&' character due to the URL Encoding.


```
device_id=82bff208d2e911dc800000163e3cf324&device_result=not+found&enabled_ck
=yes&enabled_fl=yes&enabled_js=yes&os=Windows+NT+5.1&proxy_ip=192.168.0.154&p
roxy_ip_reputation=100&screen_res=1920x1200&session_anomaly=no&summary_risk_s
core=0&time_zone=660&true_ip=220.233.95.244&true_ip_geo=AU&event_type=payment
```


Copy

URL Decoded Session Query Response

These URL Encoding has been removed from this result with each entry placed on a separate line for legibility.

```
device_id=82bff208d2e911dc800000163e3cf324
device_result=not found
enabled_ck=yes
enabled_fl=yes
enabled_js=yes
local_time_offset=263.830100
os=Windows NT 5.1
proxy_ip=192.168.0.154
proxy_ip_reputation=100
screen_res=1920x1200
session_anomaly=no
summary_risk_score=0
true_ip=220.233.95.244
true_ip_geo=AU
```

Session Query Parameters

Mandatory Parameters

The following parameters are mandatory when performing a session query. When inputting parameters, please note that any preceding and trailing whitespaces will be stripped to ensure data consistency.

Parameter Name	Parameter Type	Description
org_id	VARCHAR 8	Your org_id is the unique id associated with ThreatMetrix generated for your organization
api_key	CHAR 32	This is a unique key allocated by ThreatMetrix and associated with an **org_id.** Keep this value private.
session_id	CHAR 128	This is the session identifier included in the profiling tags for this transaction, Upper and lowercase English letters, digits, underscore or hyphen ([a-z], [A-Z], 0-9, _, -).
service_type	VARCHAR 255	Restricts which output fields are returned based on the level of access that a customer has. The service type is linked to an API Key and verified during a call. Generally, the most common service type is session-policy but may be any of the following: • **session-policy** – returns IP and device related attributes in addition to policy details. • **device** – returns only device related attributes and values.

		• **did** – returns only the device identifier. • **ip** - returns only IP related attributes and values. • **session** – returns device related attributes but no ip related attributes or policy information. • **All**- returns almost all of the above. Please note that All does not return every single attribute. • **3ds**- returns 3ds specific attributes.
event_type	VARCHAR 32 (enumerated value)	Specifies the type of transaction or event. Must be one of the following values, with the most common being: • LOGIN - A user logging into an account. • PAYMENT - Purchasing of goods/services. • ACCOUNT_CREATION - The creation of a new account. • TRANSFER -A money transfer between accounts. • TRANSACTION_OTHER - Other transaction types. • AUCTION_BID - An online auction bid. • DETAILS_CHANGE - A change of account details. • ADD_LISTING - A listing was added to an online auction site. • ACCOUNT_BALANCE - An account balance request. • TRANSACTION_HISTORY - An account transaction history was requested. • DIGITAL_DOWNLOAD - An online purchase for software goods that were delivered electronically. • DIGITAL_STREAM - An online purchase of digital media that was streamed to the user. • FAILED_LOGIN - The login attempt failed authentication. • FORGOT_USER_ID - A password/account reset. • DEPOSIT - A deposit of funds was made into the account. • LOAN_ACCEPTANCE - Represents a user accepting loan terms (making funds available). • DEVICE_REGISTRATION - Registration of a new device. • AUTH_TOKEN - Issuance of a token for user or device authentication. • PASSWORD_RESET - User performing a password reset out their account. • INIT_AUTH - An initiate authentication request • PRE_AUTHENTICATION - A pre authentication request. • BIND_POLICY - A consumer accepting an insurance policy. • GET_QUOTE - A consumer requesting an insurance quote. • CLAIM - A consumer making an insurance claim. • ADD_PAYMENT_INSTRUMENT - • MANAGE_PAYMENT_INSTRUMENT • VERIFY_PAYMENT_INSTRUMENT **NOTE:** if event_type is not specified, it will default to payment

See full list of event types.

Recommended Parameters

Parameter Name	Parameter Type	Description
policy	VARCHAR 50	Selects a policy to be used for the query. If not included, then the Policy named **default** is used. Case sensitive, spaces allowed.
account_email	VARCHAR 255	The email address associated with this event.
account_telephone	VARCHAR 255	Contact telephone number including country and city codes. All whitespace is removed. Must be in format: 0..9,<space>,(,),[,] braces must be matched.
account_first_name	VARCHAR 128	The first name of the account holder (in no account_name is provided, this is used in conjunction with the Account Last Name to provide a concatenated value for the Account Name.)
account_last_name	VARCHAR 128	The last name of the account holder (in no account_name is provided, this is used in conjunction with the Account First Name to provide a concatenated value for the Account Name.)
account_date_of_birth	TIMESTAMP 29	Account Date of Birth in YYYYMMDD format.
customer_event_type	VARCHAR 64	The sub event type that is customer specific.
transaction_id	VARCHAR 255	This is a local identifier used by the merchant to identify the transaction e.g. purchase order number.
cc_number_hash	VARCHAR (Unlimited)	The Hash of the Credit Card number. The hashing must be performed by the customer performing the API call. The customer is required to add a Salt value that is not shared with ThreatMetrix®. Recommended is SHA-256 hash (in hexadecimal format) of the Credit Card Number. Accepted currently: SHA-1 (40 bytes) SHA-224 (56 bytes), SHA-256 (64 bytes), SHA-384 (96 bytes), SHA-512 (128 bytes).
online_tld	VARCHAR 255	Description of the online_id_handle source. Used to prevent collisions between different

		online_id_handles. Warning: If online_tld is passed in an API request, you must pass an online_id_handle. Please work with an architect to determine the correct online_tld for the ID you wish to send with your query.
online_tld_handle	VARCHAR 255	A unique third-party identifier, such as a LexID or a DUNS Number. Please use online_id for rule creation and investigation. Warning: If online_id_handle is passed in an API request, you must pass an online_tld.
ach_account_hash	VARCHAR (Unlimited)	Hash of the Account Number. The hashing must be performed by the customer performing the API call. Recommended is SHA-256 hash (in hexadecimal format) of the Account Number. Accepted currently: SHA-1 (40 bytes) SHA-224 (56 bytes), SHA-256 (64 bytes), SHA-384 (96 bytes), SHA-512 (128 bytes)
drivers_license_number_hash	VARCHAR (Unlimited)	Hash of the Drivers Licence Number. The hashing must be performed by the customer performing the API call. Recommended is SHA-256 hash (in hexadecimal format) of the Drivers Licence Number. Accepted currently: SHA-1 (40 bytes) SHA-224 (56 bytes), SHA-256 (64 bytes), SHA-384 (96 bytes), SHA-512 (128 bytes)
ssn_hash	VARCHAR (Unlimited)	Hash of the Social Security Number. The hashing must be performed by the customer performing the API call. Recommended is SHA-256 hash (in hexadecimal format) of the Social Security Number. Accepted currently: SHA-1 (40 bytes) SHA-224 (56 bytes), SHA-256 (64 bytes), SHA-384 (96 bytes), SHA-512 (128 bytes)
line_of_business	VARCHAR 64	The Line of Business as specified by the customer.
application_name	VARCHAR 64	The application name that represents the application sending the event.
national_id_number	VARCHAR 64	A valid national_id_number for the national_id_type passed. If national_id_type is US_SSN, then national_id_number must be 9 digits. If national_id_type is US_SSN_HASH, then the nationl_id_number must be SHA-256 Hash of the Social Security Number.
national_id_type	VARCHAR 64	National ID type may be US_SSN or US_SSN_HASH
input_ip_address	INET (Unlimited)	IP address inputted by the customer. The IP address must be a valid external address,

otherwise user may receive the following: error_detail = input_ip_address request_result = fail_invalid_parameter

Configuring account_telephone Attribute

When the account_telephone attribute is present for an event, the following attributes are generated in addition:

- account_telephone_type
- account_telephone_is_valid
- account_telephone_is_possible

The information for these attributes requires the telephone number, country of origin of the telephone number, and the area code. The following logic is applied to determine the country of a telephone number:

- If account_telephone_geo is input from the API customer, it is used,
- If account_telephone_country_code is input from the API customer, it is used,
- If the telephone number is in E.164-compatible form, then the telephone country is taken from the telephone number. In this case, non-numeric characters such as '(', ')', ' ' and '-' are ignored. So a telephone number from the USA: "+1 (405) 234-5678" will be detected as from USA. Ensure that the "+1" and the area code is added.
- If the organisation has the "default telephone country" configured, it is used.

If the above methods are unable to determine the country of a telephone number then the attributes will display the following:

- account_telephone_type=UNKNOWN
- account_telephone_is_valid=0
- account_telephone_is_possible=0

When this occurs, the area code will be required.

Similar processing happens with all of the telephone attributes: account_telephone, primary_account_telephone, secondary_account_telephone, third_account_telephone, fourth_account_telephone, fifth_account_telephone, which have corresponding attributes for xxx_type, xxx_is_valid, xxx_is_possible.

check_circle

Recommended
When making the API call, we recommend that you pass in as many transaction related identifiers as you can such as email_address, account_login, cc_number_hash and so on. These transaction identifiers will further enhance the risk assessment for the transaction by adding to and leveraging our existing Global Intelligence Network. Passing in additional data can be integral when determining rules. For

	instance, if you pass in the transaction_amt, we are able to run rules if the amount exceeds a specific value.

Optional Parameters

Account Information Inputs

Parameter name	Parameter Type	Description
account_login	VARCHAR 255	The Account Login name
account_telephone	VARCHAR 255	Contact telephone number including country and city codes. All whitespace is removed. Must be in format: 0..9,<space>,(,),[,] braces must be matched.
account_address_street1	VARCHAR 255	The first portion of the street address component of the account address.
account_address_street2	VARCHAR 255	The second portion of the street address component of the account address.
account_address_city	VARCHAR 255	The city component of the account address.
account_address_state	VARCHAR 255	The State component of the account address.
account_address_country	VARCHAR 2	The 2 character ISO2 country code of the account address country.
account_address_zip	VARCHAR 32	The zip/postal code of the account address.
shipping_address_street1	VARCHAR 255	The first portion of the street address component of the shipping address.
shipping_address_street2	VARCHAR 255	The second portion of the street address component of the shipping address.
shipping_address_city	VARCHAR 255	The city component of the shipping address.
shipping_address_state	VARCHAR 255	The State component of the shipping address.
shipping_address_country	VARCHAR 2	The 2 character ISO2 country code of the shipping address country.
shipping_address_zip	VARCHAR 32	The zip/postal code of the shipping address.
ach_routing_number	VARCHAR 32	The routing number for the account. When using this attribute in a Condition/Expression Condition rule or Expression variable, the value will be treated as string if quotes are used to enclose it and as a number if no quotes are used.

local_attrib_1 - local_attrib_15	VARCHAR 255	These attributes can be used to pass custom attribute data that can be used by rules. When using this attribute in a Condition/Expression Condition rule or Expression variable, the value will be treated as string if quotes are used to enclose it and as a number if no quotes are used.
api_version	string	The version of the API, for example, 8.0.

Transaction Amount Inputs

Parameter name	Parameter Type	Description
transaction_amount	FLOAT8 17	A numeric currency amount. If decimal point is present, exactly two digits must follow.
transaction_currency	VARCHAR 3	The ISO 4217 currency code in which the transaction was denominated in. This field is required when transaction_amount is passed.

Attribute Reference Matrix (ARM)

The Attribute Reference Matrix is a table detailing the full set of ThreatMetrix® Attributes.

Attributes

Introduction to Attributes

An Attribute is a fundamental element in the ThreatMetrix® Platform. Attributes are used to store pieces of data that can be used by the rules engine, reporting, alerting, profiling, API and other functions of the platform.
Attributes are transmitted to the platform in the following circumstances:

Profiling

- ThreatMetrix® Web: When a device with a web browser renders a page containing the ThreatMetrix® Profiling Tags, various device related attributes are collected.
- ThreatMetrix® SDK: When a native application calls the ThreatMetrix® SDK, various device related attributes are collected.

API

- When an API call is made to our system you can pass additional attribute values. These are also referred to as 'input attributes' which are explained in more detail below.

Attribute & Portal Relationship

Each attribute has a name that represents the function of the attribute. Each attribute also has a 'friendly name' in the Portal, also known as a 'Portal Display Name'. While we attempt to ensure that both names are logical and similar, there may occasionally be differences between the two.
An example of differing Attribute and Portal Display names is the 'fuzzy_device_id' attribute, which is displayed as 'ThreatMetrix® SmartID®' in the portal.

The Attribute Reference Matrix displays a correlation of each attribute and its Portal Display Name, as well a description, type, length, input/output type and other useful informational elements.

Attribute Visibility in the Portal

The majority of attributes available are enabled and visible by default in the Portal. However, some attributes may not appear until they are manually enabled by customers. To do this, please login to the Portal, select the Settings tab from the top navigation menu, then select the Attributes tab from the sub-menu tabs. Ensure all attributes you wish to be visible are checked, then click the Submit button at the bottom left of the screen.

> info
>
> **Note**
>
> You may need to logout and login again after enabling/disabling attributes for the changes to take effect.

Input & Output Attributes

The terminology used to describe how the data is enumerated in an attribute is referred to as Input and Output.

Attributes collected by the profiling procedure are referred to 'Output Attributes' because they are simply outputted by the system into their respective Attributes. They cannot be enumerated with user-specified input via the API. 'IP Address' and 'Screen Resolution' are both examples of output attributes.

Customers can pass (or Input) attribute values when performing the API call. Attributes passed by the API procedure are referred to 'Input Attributes' because they are enumerated by user-specified input. 'Email Address' and 'Credit Card Hash' are both examples of input attributes. It is highly recommended to pass as many relevant input attributes as possible. Providing more input attributes at the time the API call is performed enables the rules engine to evaluate more data points, and as such, make a stronger decision about the transaction. Personally Identifiable Information (PII) that is passed at API time is handled in a secure manner to ensure privacy compliance.

The Attribute Reference Matrix displays a correlation of each attribute and its Portal Display Name, as well a description, type, length, input/output type and other useful informational elements.

Please note that the API does not accept <, > or % characters. These are simply ignored and removed from the Input.

Attribute Types

There are various types and categories of attributes available in the ThreatMetrix® Platform. The Attribute Reference Matrix has a 'Category' and 'Type' column that allows you to easily filter and sort by the various Attribute Types and Categories.

The ThreatMetrix® platform has two distinct types of data elements: **Entities** and **Standard Attributes**. ThreatMetrix® strictly controls and defines the data types and validation of each data element. Customers do not have this capability.

Each Attribute Category and Type is explained below:

Entities

Entities can be thought of as collections of attributes or 'top level' attributes. An example of an Entity is 'true_ip' (the True IP Address of the connecting device). Each Entity has numerous corresponding Entity Attributes (described below) that comprise the collection. These Entity Attributes provide more information about the Entity such as the date/time it was first seen or the last time it was updated. Every time an Entity is collected (either by the device based attributes during profiling, or the customer supplied data via the Session Query API) the corresponding Entity Attributes are automatically generated.

Entity Time to Live (TTL)

Entities have a TTL period of **180 days** but will continue to live in the system as long as they are queried at least once in that period.

Please note that cookies are preserved for a period of 450 days ensuring that the same ThreatMetrix® ExactID® and ThreatMetrix® SmartID® are returned within the 450 day period. However, the entities themselves are retained for 180 days.

Entity Attribute

Entity Attributes are a set of attributes that belong to an Entity. These are automatically generated by the platform when the Entities themselves are stored. They are effectively metadata that provide more information about an Entity. An example of an Entity Attribute is 'true_ip_first_seen'. The name of each Entity Attribute is preceded by the entity's name and an underscore. In the above example 'true_ip' is the entity, and '_first_seen' is the Entity Attribute.

Every Entity has a similar set of corresponding Entity Attributes (e.g _first_seen, _last_update etc..) although some attributes, such as the Address and IP related attributes, have some additional ones such as _city and _country. A full list of Entity Attributes is below:

- **_activities** - the list of activities such as Login, Payment or Transfer associated with the entity based off its assertion history. Also contains the local trust tags.
- **_assert_history** - the assertion history for the entity based on the assertion confidence such as "Confirmed Bad" or "Suspicious".
- **_attributes** - the industry type associated with the entity activity. This attribute also displays any global trust tags configured for the entity.
- **_first_seen**- the date the entity was first encountered.
- **_last_assertion** - the date of last assertion on that entity.
- **_last_event** - the date of the last event the entity was involved in.
- **_last_update** - the date the entity was last updated.
- **_result** - indicates whether this entity has been encountered before or for the very first time.
- **_score** - the risk score for this entity between +100 (good) and -100 .
- **_worst_score** - the lowest score ever reached by this entity between 0 (neutral) and -100 (bad) .

The whole purpose of generating these Entity Attributes is to provide customers with a greater level of detail regarding the current status and history of an Entity.
For more detail on the Entity Attributes please view the Attribute Reference Matrix. You can sort/filter the 'Attribute Type / Category' column to easily locate the Entity Attributes.

Standard Attributes

Not all attributes in the ThreatMetrix® Platform are Entities or Entity Attributes. These remaining attributes are simply referred to as Standard Attributes. An example of a Standard Attribute is 'screen_resolution'. Entity Attributes ARE NOT generated for Standard Attributes. For example, as there is no real value in understanding the first time a screen resolution was seen, additional Entity Attributes are not generated.

Attribute Categories

For reference, every attribute in the ThreatMetrix Platform belongs to a category. These categories are listed in the Attribute Reference Matrix. A description of each category is below.

Session Attribute

Session Attributes contain information regarding an individual session. A session is defined as any occurrence of a device either rendering our ThreatMetrix Cloud profiling tags or being profiled via our ThreatMetrix Mobile library. This commonly includes logins, account creations and transactions. These are generated automatically by our profiling service each time a session occurs and are primarily used to identify the session and provide additional information regarding the session. Examples of Session Attributes are 'session_id' and 'datetime'.

NOTE: The session_id attribute is **not** automatically generated by our system. It has to be generated by your web application and is used to uniquely identify the session/transaction. Please see Session ID for more information.

Device Attribute

Device Attributes contain identifying information about the device that performs a session. These aid the rules engine in making more informed decisions as to whether the device is known or unknown to our platform and help identify any device anomalies or irregular behavior. Examples of Device Attributes are 'device_id' and 'screen_res'.

Page Fingerprint Attribute

Page Fingerprint Attributes relate specifically to the Page Fingerprinting technology that identifies webpages that have been modified by malware, either by a Man in the Browser infection locally on a device or a compromised web server. Examples of Page Fingerprint Attributes include 'page_id' and 'page_fingerprint_summary'.

Third Party Attribute

Third Party Attributes are used to communicate with external Third Party services, such as address and phone number verification services. Additional third party services will be added as they become available to provide further intelligence to transaction analysis. Examples of Third Party Attributes include 'tps_result' and 'tps_error'.

True IP Attribute

True IP Attributes contain identifying information about the 'true IP address' that was detected during a transaction. A True IP is an IP address that was determined to be the REAL originating IP address of a device, as opposed to the Proxy IP address that was presented by the device. Our Proxy Piercing feature allows the True IP to be captured. Examples of True IP Attributes include 'true_ip_city' and 'true_ip_latitude'.

Proxy IP Attribute

Proxy IP Attributes contain identifying information about the proxy server that was used (if any) during a session. Examples of Proxy IP Attributes include 'proxy_ip' and 'proxy_ip_isp'.

Browser Attribute

Browser Attributes contain information about the browser that was used during a session. This includes useful information such as the type of browser and language of the browser, and whether the browser has various plugins enabled, such as JavaScript, and Silverlight. Examples of Browser Attributes include 'browser_language' and 'http_referer'.

Payment Attribute

Payment Attributes store information about a session where a payment or transaction was performed. These are typically handed to our platform by customers at the time the API call is made. It is highly recommended to pass us Payment Attributes whenever possible as it enables our rules engine to apply payment specific logic and help reduce fraud by seamlessly authorizing good transactions and scrutinizing suspicious transactions. Examples of Payment Attributes include 'cc_number_hash' and 'transaction_amount'.

Identity Attribute

Identity Attributes store information about a user's identity that performed the session. These are typically handed to our platform by customers at the time the API call is made to provide additional identification elements. It is highly recommended to pass us Identity Attributes whenever possible as it enables our rules engine to apply identity specific logic and help reduce fraud by seamlessly authorizing good logins and scrutinizing suspicious logins. Examples of Identity Attributes are 'ssn_hash' and 'drivers_license_number_hash'.

Account Attribute

Account Attributes store information about the account that was used in a session. These are typically handed to our platform by customers at the time the API call is made. It is highly recommended to pass us Account Attributes whenever possible as it enables our rules engine to apply account specific logic and help reduce fraud by seamlessly authorizing good transactions/logins and scrutinizing suspicious transactions/logins. Examples of Account Attributes include 'account_login' and 'password_hash'.

Custom Attribute

Custom Attributes can be used by customers to pass custom attribute data as strings that can be processed by the rules engine. For example, a retailer could provide the SKU of a purchased item from their online store as a custom attribute. The contents of these attributes will automatically be encrypted by our platform so only customers can see the decrypted values and ThreatMetrix® can only see the encrypted values. There are a number of custom attributes available (continually updated) named as follows:

- **Portal Display Name:** Custom Attribute 1, Custom Attribute 2, etc..
- **Attribute Name:** local_attrib_1, local_attrib_2, etc..

Custom Match Attribute

Custom Match Attributes are used to store a hashed result of the Multiple Attribute Velocity rule. To better comprehend the Custom Match Attributes, it helps to understand how the Multiple Attribute Velocity rules function: they enable customers to specify up to 12 different attributes as well as a numerical threshold. When the rules are processed, if the specified attributes have been seen together equal to or greater than the specified threshold then the rule will trigger, create a MD5 hash of all of the combined attributes and store the resulting hash in a Custom Match Attribute specified by the customer. You can then use the Attribute Anomaly rule to detect situations where there are variances in the combined set of attributes. There are 8 Custom Match Attributes available, numbered from 1-8, named as follows:

- **Portal Display Name:** Custom Match ID 1, Custom Match ID 2, etc..
- **Attribute Name:** custom_match_1, custom_match_2, etc..

Custom Number Attribute

Custom Number Attributes are used to store the numerical threshold result value outputted by the following rules: Attribute Anomaly, Attribute Velocity, Entity Velocity, Event Type Velocity, Distance Traveled, Multiple Attribute Velocity, and Transaction Sum Exceeds Threshold. As an example, if the Multiple Attribute Velocity rule (explained above in the Custom Match Attributes) has a configured threshold of 3, and the rule detected that the attributes have historically matched 7 times, then a value of 7 would be stored in the Custom Number Attribute specified in the rule. There are 4 Custom Number Attributes, numbered from 1-4, named as follows:

- **Portal Display Name:** Custom Number 1, Custom Number 2, etc..
- **Attribute Name:** custom_count_1, custom_count_2, etc..

Conditional Attribute

Conditional Attributes can be used by customers to pass custom attribute data that can be used by the Condition rule and the Attribute in List Rule. These attributes are encrypted and should be used whenever personally identifiable information (PII) needs to be passed. There are a number of Conditional Attributes (more may be added), named as follows:

- **Portal Display Name:** Conditional Attribute 1, Conditional Attribute 2, etc..
- **Attribute Name:** condition_attrib_1, condition_attrib_2, etc..

Unencrypted Conditional Attribute

Conditional Attributes can be used by customers to pass custom attribute data that can be used by the Condition rule and the Attribute in List Rule. These attributes are **NOT** encrypted and should not be used to pass personally identifiable information (PII). There are 2 Unencrypted Conditional Attributes, numbered from 1-2, named as follows:

- **Portal Display Name:** Unencrypted Conditional Attribute 1, Unencrypted Conditional Attribute 2
- **Attribute Name:** unencrypted_condition_attrib_1, unencrypted_condition_attrib_2

Entity Reputation Score for attributes

ThreatMetrix® has implemented an Entity Reputation Score for every entity tracked within the Global Network.
The score is hugely beneficial as it can be used in Conditional rules as additional inputs into the overall risk of the transaction. The entity scores (such as IP, Device ID, etc) are calculated based on composite scores across our network. The algorithm is proprietary, and will be continually adjusted to improve it. In general, a high rate of negative or positive transaction scores associated with an entity will alter its score, and the weighting will be adjusted so that as the entities are seen across multiple org_ids, their weighting is increased. Read more about it in Risk Weighting (Entity Reputation Score) for Global Rules.

List of Rules affected by the Risk Weighting include:

- Device in Global List
- IP in Global List
- IP in Global Threat List
- Proxy IP in Global Threat List

Implementation Overview

API Key

An API Key is a string of characters that is used to authenticate the API Calls made by a customer to the ThreatMetrix Platform. All ThreatMetrix API requests are authenticated using the ThreatMetrix Dynamic Decision Platform (DDP) API Security. Each customer is issued one API Key when first acquiring solutions from ThreatMetrix. An API Key is unique and associated with an Org ID. If a customer has multiple Org IDs, E.g. one for testing an one for production then each Org ID will have a differing API Key.

warning

Warning
It is the customer's responsibility to protect this API Key and not disclose it to any unauthorized third party. If you suspect your API Key has been compromised please contact Technical Support and you will be issued a new API Key.

An example API Key is: 325euk66x0lckl7j

Architectural Overview

The following steps and diagram represent the typical ThreatMetrix Web and Mobile solutions flow:

1. ThreatMetrix Profiling

ThreatMetrix profiles the user's device. The profiling occurs differently depending on the ThreatMetrix product in use:

- **ThreatMetrix Web**:
  - The customer places ThreatMetrix **Profiling Tags** on the desired webpages (commonly *Login*, *Payment*, and *New Account Creation*).
  - The user visits a customer's webpage containing the Profiling Tags via a web browser on a device. This could be a computer, tablet, smartphone, or any other device with a web browser.
  - The web browser will execute the Profiling Tags, causing the browser to collect device related attributes.
- **ThreatMetrix SDK**:
  - The customer includes the ThreatMetrix SDK library in their native applications.
  - The user launches the customer's native application.
  - When the library is invoked it will perform profiling.

2. Session ID and Attribute Transmission

The user's device transmits the **Session ID** and the device related attributes to the ThreatMetrix Platform.

3. Customer Backend Transmission

The user completes their browser or application session and the **Session ID** is transmitted to the customer's backend — along with the session data that the application usually receives (e.g., *username*, *password hash*, *transaction value*, etc.)

4. Session Query

The customer's backend performs a **Session Query** via the ThreatMetrix API which triggers the **Rules** engine to perform a configurable analysis of the session. The ThreatMetrix platform returns detailed information regarding the session, enabling the customer's backend to take action based on the outcome of the executed rules.

Implementation Resources

The following resources are required when performing a typical implementation of ThreatMetrix solutions. Please review the list below to ensure you understand and have obtained each required resource prior to performing an implementation.

Once you have met the below requirements, please view the Implementation Steps article for guidance on how to begin your deployment.

#	Name	Required for...	Obtained from...	Additional Information
1	Portal Credentials	Accessing the ThreatMetrix Portal	Automatically generated and delivered to you via Email when ThreatMetrix Services creates a Portal Account for your organization. The ThreatMetrix Portal allows customers to: • Review events • Manage policies used to detect fraud and identify good customers • Analyze potential fraud and malware activity and take action • Configure email alerts and notifications	Introduction to the Portal
2	Org ID	Profiling, APIs	Automatically generated when ThreatMetrix Services creates a Portal Account for your organization. Available in the Portal via the Admin > API Keys tab.	Org ID
3	API Key	APIs	Automatically generated when ThreatMetrix Services creates a Portal Account for your organization. Available in the Portal via the Admin > API Keys tab.	API Key
4	Session ID	Profiling, APIs	ThreatMetrix Web: Generated by the customer's web application. ThreatMetrix SDK: Generated by the customer's application or by the ThreatMetrix SDK.	Session ID
5	SSL Certificate for Enhanced Profiling	Profiling	Work with the ThreatMetrix Services team to implement enhanced profiling.	Enhanced Profiling via Hosted SSL

(Mandatory for ThreatMetrix SDK)

Org ID

An Organization ID, commonly referred to as an Org ID, is a string of characters that serves to uniquely identify a ThreatMetrix customer when performing Profiling and API Calls. When an Org ID is created it is also assigned a unique public/private keypair in our backend cryptography engine that is used to encrypt all Personally Identifiable Information (PII) that customers choose to transmit to us as part of the API Call. E.g. Credit Card Hashes, Street Addresses etc. The methodology of Org IDs and associated unique keypairs ensures customer data is secure and isolated in our platform.

An example Org ID is: sfqa1ul2

If you do not have an Org ID or your Org ID is not working please contact our Support team.

Session ID

The Session ID is a temporary identifier that is unique to the visitor's session. It is the primary element used to identify and query end-user sessions on customer websites. Both ThreatMetrix Cloud and ThreatMetrix Mobile require a Session ID to be present during a session as detailed below:

ThreatMetrix Web

The Session ID must be a unique ThreatMetrix Session ID to be included within the Profiling Tags in order for our service to uniquely identify the end user's session. The Session ID must be generated by the customer's web application.

Generating a Session ID

Some recommended methods of generating a Session ID are:

- Referencing an existing session identifier from a cookie and appending the date and time in milliseconds to the end of the identifier and then applying a hexadecimal hash to the concatenated value to produce a completely unique Session ID.
- Referencing an existing session identifier from the web application and applying a hexadecimal hash to the value to obfuscate the identifier.
- Generate a completely arbitrary, random number.

Inserting the Session ID into the Profiling Tags

Once the Session ID has been generated it needs to be inserted into the profiling tags in the session_id= field as in the following example:

```
session_id=16a9508e007c82f33572ac0907f1ec1b
```

> **Note**
> NOTE: A full example of the Profiling Tags can be seen in the Profiling Tags documentation. Note the session_id= parameter.

ThreatMetrix SDK

Please refer to the Using ThreatMetrix Mobile article for detailed information.

Session ID Format

The Session Id may be up to 128 bytes long and must consist of the following characters only:

- Upper and lowercase English letters (a-z, A-Z)
- Digits (0-9)
- Underscore (_)
- Hyphen (-)

Session ID Length

session_ids will need to be at least 10 valid characters long, or an empty JS response, 204 (OK) will be returned. Existing organizations may now request to mandate a minimum length of 10 for a session_id.

Minimum length is after sanitization (removal of non-conformant characters).
If the session_id exceeds 128 characters, it will be truncated to 128 characters in the API response.

Session ID Validation

The validation of the session_id is dependent on whether the Check Session ID flag is set to true or not for your organization. This flag is set to true by default for all new customer organizations. When this flag is set to true, the session_ids will need to be at least 10 valid characters long and no invalid characters must be present. If the session_id does not comply, the profiling call will fail with a 204 (OK) response and the API response will return the request_result as "fail_invalid_parameter" and the error_detail as "session_id".

If this flag is not set, then the minimum length will not matter and any non-conformant characters will be sanitized and the call allowed to proceed.

check_circle

Recommended
An MD5 or SHA hashed value will conform to the above format requirements.

Session ID Notes and Limitations

- Ideally, each profiling session should use a unique Session ID.
- Multiple API calls may be performed with the same Session ID as long as no new profiling was done, as the calls will access the same session data.
- *handle_id* is a deprecated attribute that has been superseded by *session_id*.

Profiling

Introduction to Profiling

Profiling is a mechanism by which ThreatMetrix® collects information about the device that is accessing a customer's online service, either via a website or a native application. Information about the profiled device is collected and stored as Attributes which are evaluated by ThreatMetrix® when performing a risk/fraud assessment.

How does it work?

Profiling is performed using different methods depending on the ThreatMetrix® solution in use. Profiling for these solutions are described below.

ThreatMetrix Web

ThreatMetrix Web allows customers to integrate ThreatMetrix with their **webpages**.

Profiling is performed via the ThreatMetrix profiling tags. This is a JavaScript tag placed on relevant customer pages. When a visitor accesses these page via their browser, the profiling tags will cause the device to communicate with the ThreatMetrix platform and transmit various device attributes. Profiling tags have no visual representation and has minimal impact on page load times. The profiling process takes 2-5 seconds to complete (depending on the network bandwidth). The ThreatMetrix Profiling Tag creates an iframe when loaded which asynchronously performs profiling. This means that page load times are fast, and there is no interference with existing page load routines.

ThreatMetrix SDK

ThreatMetrix SDK allows customers to integrate ThreatMetrix with their **native applications**.

TrustDefender SDK is embedded into the native application. The customer then invokes the SDK at the required times in their application workflow. When the SDK is invoked, it will cause the device to communicate with the ThreatMetrix platform and transmit various device attributes. The ThreatMetrix SDK has no visual representation and has minimum impact on app performance. The profiling process takes 2-5 seconds to complete (depending on the network bandwidth and processing power of the device). The execution of the SDK is performed asynchronously and will not interfere with the app.

Basic Profiling Integration

This section outlines the basic method for integration profiling into a web page. This method does not support the advanced features of adblock defense, re-profiling or support for Single Page Applications. For these features, please use Advanced Profiling Integration.

Profiling Tag Code Sample

Tag Parameters

There are various parameters that are specified in the profiling tag:

- **org_id** (mandatory): a string of characters that serves to uniquely identify a ThreatMetrix customer. See the Org ID article for more information.
- **session_id** (mandatory): a unique identifier that identifies this one event performed by the user. See the Session ID article for more information.
- **page_id** (optional): this is used to identify specific pages to the ThreatMetrix Platform to support the Man in the Browser detection functionality. See the MitB documentation for more information.
- **Custom Profiling Domain** (optional): this is the domain that serves the JavaScript tag. The default is h.online-metrix.net. Customers can optionally implement the Enhanced Profiling feature whereby ThreatMetrix hosts an SSL certificate on behalf of the customer. The result is that the profiling domain then points to a hostname that appears to be hosted by the customer, and not ThreatMetrix. This has several benefits. See the Enhanced Profiling Documentation for more information.

Supported Tag Deployments

There are various methods available when deploying the profiling tag, detailed below.

Basic <script> Tag with <noscript> (Recommended)

This is the recommended deployment. This includes a basic script tag which loads a Javascript resource: tags.js, as well as an additional noscript tag. The purpose of the noscript tag is to ensure that profiling occurs, even if JavaScript is disabled.

Tag Placement

- This basic script tag should be placed in the head tag for optimal performance.
- The noscript tag must be placed in the body tag, as in the below example. The noscript tag **must not** be placed in the head tag as it contains an iframe tag. Placing iframes in the head tag violates W3C validation and may cause issues.

```
<head>
    <script type="text/javascript" src="https://h.online-metrix.net/fp/tags.js?org_id=orgid123&session_id=01f50c4d1430a620a3b50005ffe98541"></script>
</head>
<body>
    <noscript>
        <iframe style="width: 100px; height: 100px; border: 0; position: absolute; top: -5000px;" src="https://h.online-metrix.net/fp/tags?org_id=orgid123&session_id=01f50c4d1430a620a3b50005ffe98541"></iframe>
    </noscript>
</body>
```

Basic script Tag

This includes a single `<script>` tag that loads a JavaScript resource: `tags.js`. If JavaScript is disabled, this tag will not load and profiling will not occur.

Tag Placement

This basic script tag should be placed in the head tag for optimal performance.

```
<head>
    <script type="text/javascript" src="https://h.online-metrix.net/fp/tags.js?org_id=orgid123&session_id=01f50c4d1430a620a3b50005ffe98541"></script>
</head>
```

Session ID

When re-using session_ids, only the most recent profile event is used for the API call. If session_ids are re-used across devices, then the API might not be using the profiling data intended.
Please read the Session ID article for more information about the Session ID format, length and limitations.

Advanced Profiling Integration

This section outlines how to use the Profiling Javascript Toolkit to implement profiling rather than using the Basic Profiling Integration (tags.js script).
The benefits of using this profiling integration are:

- Better defense against adblockers with URL encryption.
- Customer ability to re-profile by calling the profile() function multiple times.
- Reduction of unknown sessions due to adblocked profiling domain.
- Better support for Single-Page applications

Note: Advanced Profiling Integration is required for SPA (Single Page Application) implementation.

Installing the snippet and invoking profiling

> info
>
> **Note**
> Please note that this toolkit is updated from time to time and we recommend that customers download the latest version.

Latest Release: fp-clientlib-v5.js (Version 5)
Release Date: May 18, 2021
Download the latest version: fp-clientlib-v5.js

1. Download the toolkit. Customers should migrate to the new version of the library and are advised to host this JS file on their own websites.
2. Rename the downloaded file into something unique, E.g., "company_toolkit.js"
3. Place the renamed file in a publicly accessible location on your web server.
4. Include a reference to the toolkit on the page you wish to have profiled:
5.
```
<script type="text/javascript"
src="path/to/company_toolkit.js"></script>
```

   Copy

6. Trigger profiling asynchronously by calling it in a script on your page and including the appropriate parameters:

Please ensure that the function 'threatmetrix.profile' is renamed; this function is part of the toolkit customers download when implementing tag obfuscation and should be renamed or randomized so that it does not call attention to the fact that ThreatMetrix is being used.

```
<script type="text/javascript">threatmetrix.profile(profiling_domain, org_id,
session_id, page_id);</script>
```

Copy

Substitute profiling_domain, org_id, session_id and optionally, page_id with the appropriate parameters. Note that these should all be passed as string types.

Note that you will need to generate session_id on your server-side in order to reference the session query from your back end.

Profiling domain will be "h.online-metrix.net" if you have not setup [Enhanced Profiling via self-hosted SSL](#).

For example:

```
<script type="text/javascript">threatmetrix.profile("h.online-metrix.net",
"abcd1234", "01f50c4d1430a620a3b50005ffe98541");</script>
```

Copy

Example (JSP)

An example of how the client toolkit could be used on a single-page JSP application :

```
<head>
<script type="text/javascript" src="path/to/toolkit.js"/>
</script>
<!-- other head content -->
</head>
<body>
<script type="text/javascript">
    var session_id = <%=generateSessionId()%>;
    threatmetrix.profile("<profiling_domain>", "<org_id>", session_id);
</script>
<noscript>
```

```
        <iframe style="width: 100px; height: 100px; border: 0; position:
absolute; top: -5000px;" src="<profiling
doman>/fp/tags?org_id=<orgid123>&session_id=<session_id>"></iframe>
</noscript>
</body>
```

Profiling Tag Code Sample

Tag Parameters

There are various parameters that are specified in the profiling tag:

- **org_id** (mandatory): a string of characters that serves to uniquely identify a ThreatMetrix customer. See the Org ID article for more information.
- **session_id** (mandatory): a unique identifier that identifies this one event performed by the user. See the Session ID article for more information.
- **page_id** (optional): this is used to identify specific pages to the ThreatMetrix Platform to support the Man in the Browser detection functionality. See the MitB documentation for more information.
- **Custom Profiling Domain** (optional): this is the domain that serves the JavaScript tag. The default is h.online-metrix.net. Customers can optionally implement the Enhanced Profiling feature whereby ThreatMetrix hosts an SSL certificate on behalf of the customer. The result is that the profiling domain then points to a hostname that appears to be hosted by the customer, and not ThreatMetrix. This has several benefits. See the Enhanced Profiling Documentation for more information.

Supported Tag Deployments

There are various methods available when deploying the profiling tag, detailed below.

Basic <script> Tag with <noscript> (Recommended)

This is the recommended deployment. This includes a basic `<script>` tag which loads a Javascript resource: `tags.js`, as well as an additional `<noscript>` tag. The purpose of the `<noscript>` tag is to ensure that profiling occurs, even if JavaScript is disabled.

Tag Placement

- This basic `<script>` tag should be placed in the `<head>` tag for optimal performance.
- The `<noscript>` tag must be placed in the `<body>` tag, as in the below example. The `<noscript>` tag **must not** be placed in the `<head>` tag as it contains an `<iframe>` tag. Placing iframes in the head tag violates W3C validation and may cause issues.

```
<head>
<script type="text/javascript" src="path/to/toolkit.js"/>
</script>
<!-- other head content -->
```

```
</head>
<body>
<script type="text/javascript">
    var session_id = <%=generateSessionId()%>;
    threatmetrix.profile("<profiling_domain>", "<org_id>", session_id);
</script>
<noscript>
        <iframe style="width: 100px; height: 100px; border: 0; position:
absolute; top: -5000px;" src="<profiling
doman>/fp/tags?org_id=<orgid123>&session_id=<session_id>"></iframe>
</noscript>
</body>
```

Basic <script> Tag

This includes a single `<script>` tag that loads a JavaScript resource: `tags.js`. If JavaScript is disabled, this tag will not load and profiling will not occur.

Tag Placement

This basic `<script>` tag should be placed in the `<head>` tag for optimal performance.

```
<head>
    <script type="text/javascript" src="https://h.online-
metrix.net/fp/tags.js?org_id=orgid123&session_id=01f50c4d1430a620a3b50005ffe9
8541"></script>
</head>
```

Session ID

Please read the Session ID article for more information about the Session ID format, length and limitations.

Previously, when a session_id was reused between different profiling events on different devices, profiling results simply overwrite previous results. This could lead to 'intermingled' results where session_ids were heavily reused. We now strictly split results between profiling events so these results are not mixed, however we are not able to fully do this where profiling has been implemented directly with clear.png and check.js javascript calls.

In order to avoid issues with session_id reuse, customers should make sure they implement profiling on their web pages using tags.js, or better still, the javascript toolkit as per Tag Obfuscation and Ad-blocker Avoidance.

Where previously, using a direct check.js profiling implementation and re-using session_ids could lead to results shared between events and device_id sharing and so on, from 11.2 we will be dropping data that we are unable to strongly associate with a particular profiling session. For the most part, this would be the flash results where the flash tags are directly implemented on a webpage in a check.js implementation. Note that flash results are only a small part of device identification and this is not expected to cause significant issues.

Customers in this situation should update their profiling implementations to use tags.js or the above javascript toolkit for profiling instead.

Please Note:

Flash has been removed from profiling service for all customers as of v12.3 on June 27, 2021.

ThreatMetrix SDK

Introduction to ThreatMetrix SDK and FAQ

ThreatMetrix separates Software Development Kits (SDKs) into Android, iOS, OSX, Windows, and Java. Follow the information below to get and use a ThreatMetrix SDK.

Using ThreatMetrix SDK

Downloading ThreatMetrix SDK

ThreatMetrix SDKs are available under **SDK Downloads**. Open the appropriate SDK and download the file.

In order to access the **Downloads** section, your company must be in contract with ThreatMetrix. Please contact your Account Manager, or contact the support team: tmx.support@lexisnexisrisk.com

Once you have determined that you have the necessary support level in place, you may contact ThreatMetrix Support and request to gain access to the ThreatMetrix SDK and the Downloads sections which is located within the KnowledgeBase. A member of the ThreatMetrix support staff will then add the privilege to your user account, after which, you will be able to access these sections upon logging in.

Required Operating System Permissions

Required operating system permissions are introduced as a result of implementing ThreatMetrix SDK. A major design goal is that wherever possible, additional permissions are **optional**, instead of making them **mandatory**.

The Getting Started and Project Configuration article for each respective platform describes the process of specifying the below permissions during development.
Permissions are described below:

Android

Mandatory Permissions
The application must include the following permission in its manifest file:

```
<uses-permission android:name=
"android.permission.INTERNET"></uses-permission>
```

Optional Permissions

1. Accessing WIFI State
Including this permission in their application allows the gathering of precise information of the active WIFI network (SSID, BSSID and RSSI).

```
<uses-permission android:name=
"android.permission.ACCESS_WIFI_STATE" ></uses-permission>
```

2. Accessing Fine and Coarse Locations

Customers can use Location Services if they have one of the below mentioned permissions. Gathering fine location information is highly recommended as it enables us to detect location updates in shorter distances.

```
<uses-permission android:name=
"android.permission.ACCESS_FINE_LOCATION" ></uses-permission>
```

OR

```
<uses-permission android:name=
"android.permission.ACCESS_COARSE_LOCATION"></uses-permission>
```

iOS

If using LocationServices, then, the application **MUST** specify the message used during the permission prompt by adding an entry to the application .plist file:

```
<key>NSLocationWhenInUseUsageDescription</key>
<string>Insert message here</string>
```

Viewing ThreatMetrix SDK Logs

Android

The Android logging system provides a mechanism for collecting and viewing system debug output. Logcat dumps a log of system messages, which include things such as stack traces when the emulator throws an error and messages that you have written from your application by using the Log class. You can run Logcat through ADB or from DDMS, which allows you to read the messages in real time.
For more information, please view the following external articles:

- Android Logging: http://developer.android.com/tools/debugging/debugging-log.html
- Android Logcat: http://developer.android.com/tools/help/logcat.html

In AndroidStudio

- Select your application
- Select the log level

In Logcat command line

- add -v threadtime to print time, process id, and thread id
- run the grep command for SDK tag "t.c.tdm"

Please note that some devices like Huawei filter all logs, warnings and errors. Hence, enabling debug logs on these type of devices is based on the specific model and may require you to search for specific scenarios on the internet.

iOS

Please see the following article for iOS logging and debugging information: http://developer.apple.com/library/ios/#technotes/tn2151/_index.html

Using an SSL Certificate to Communicate to the ThreatMetrix Profiling Server

Customers are required to use the ThreatMetrix Enhanced Profiling Service for all ThreatMetrix SDK Implementations. This involves acquiring an SSL certificate as described in Enhanced Profiling via Hosted SSL.

The reasons for this requirement are:

- Secures the traffic between the ThreatMetrix SDK and the ThreatMetrix Profiling Server
- Makes it more difficult for fraudsters to identify that data is being routed to a third party from the native app
- Enables ThreatMetrix to uniquely identify customer specific profiling traffic originating from the ThreatMetrix SDK
- Enables ThreatMetrix to throttle or disable traffic originating from compromised ThreatMetrix SDK instances
- Enables ThreatMetrix to allocate additional resources to specific customer implementations with high profiling traffic volume

Xcode Compatibility for iOS
Apple often requires new Xcode builds for app submission. This can introduce compatibility issues so we recommend you review the compatibility matrix listed on the Apple website: Xcode - Support - Apple Developer

ThreatMetrix SDK Support Lifecycle Policy

The ThreatMetrix SDK Support Lifecycle Policy provides consistent and predictable guidelines for the availability of support.

ThreatMetrix releases major SDK versions about once every year and minor versions as the need arises. The software product version will show as (Major).(Minor) (e.g., 5.1).

The SDK is fully supported and maintained until the *End of Maintenance (EOM)* date in accordance with the following conditions:

- ThreatMetrix will provide code-level maintenance in the form of bug fixes, work-arounds, maintenance releases, or patches.
    - Bugs must be reproducible by the customer's app development/support team before any fix request is submitted.
- ThreatMetrix will determine the best way to resolve reported issues which will commonly require an upgrade to the latest release, as full support is only for the latest major version.
    - Any patch to a major version will require an upgrade to the latest minor version as well.
    - Fixes for critical security issues observed as a result of penetration testing or recently discovered vulnerabilities can be extended to one prior major version.
- Technical support will continue to be available and provided during this phase as long as a valid support contract exists and is paid in full.

The EOM Period for a Major Software release, “N,” starts when the N+1 release reaches *General Availability (GA)* (where the product is available for Sale, Support, and Maintenance). The End of

Maintenance (EOM) period of a Major Software release is the lessor of: two (2) years from General Availability or one (1) year after the N+1 version reaches General Availability.

Once *End of Life (EOL)* is reached, the following applies:

- The SDK may continue to operate and may be used on an “as-is” basis within the terms of licensing agreements, but future operation is not guaranteed.
- Technical support and software updates will no longer be available once a product has reached its End of Life date.
- Historical information in the Knowledgebase or other online resources may still be present, but it is no longer updated and is provided on an “as-is” basis.

The EOL period of a Major Software release is the lessor of: three (3) years from the date it first became Generally Available or two (2) year after the N+1 version becomes Generally Available. Access to ThreatMetrix’s technical support team will be available until a product reaches its End-of-Life date provided that an active support contract exists.

Frequently Asked Questions (FAQ)

Support Questions and Errors

Where can I get support for a ThreatMetrix SDK?

Please email tmx.support@lexisnexisrisk.com and please include the following information:

1. Confirm with ThreatMetrix that your Organization (Org ID) has been activated for ThreatMetrix SDK. Profiling calls will fail if this has not been enabled.
2. Environment information:
    1. Which Platform?
        1. iOS or Android?
        2. Did this issue come from an emulator or simulator?
    2. Which version of the operating system?
        1. Which IDE?
        2. XCode or Eclipse, Java Beans, Android Studio etc.
    3. Which version(s) of TheatMetrix SDK?
3. Detailed description of the issue you are having.
4. Your expected result versus your actual result.
5. All debug log files and stack traces generated if the app is crashing
6. Any other custom fields you are passing
7. Are you passing a custom session ID or using the one automatically generated by ThreatMetrix SDK?

It is best practice to use the most recent release of any and all software, whether it be testing environments, OS version, or ThreatMetrix SDK version.

What is the error "HTTP Load Failed" / SSL Handshake errors on iOS 9?

Customers implementing ThreatMetrix SDK on iOS 9 may experience errors relating to SSL Handshake between the SDK and the ThreatMetrix profiling servers. Example errors include:

```
CFNetwork SSLHandshake failed (-9824)
NSURLSession/NSURLConnection HTTP load failed (kCFStreamErrorDomainSSL, -
9824)
```

Apple implemented changes requiring stronger SSL ciphers in iOS 9, as documented in the iOS 9 release notes in the **Networking**chapter:
https://developer.apple.com/library/prerelease/ios/releasenotes/General/RN-iOSSDK-9.0/

Customers using ThreatMetrix SDK that are referencing the old profiling infrastructure either via our generic hostname (h.online-metrix.net) or via Enhanced Profiling via Hosted SSL(whereby ThreatMetrix hosts an SSL certificate on behalf of the customer) will be affected. ThreatMetrix has responded by offering new profiling servers that meet these requirements.
This has a different impact depending on the version of ThreatMetrix SDK you are using:

ThreatMetrix SDK v1.x - v3.0x

Customers must manually specify the following profiling hostname:
`h-sdk.online-metrix.net`
NOTE: Customer running these earlier versions of ThreatMetrix SDK are strongly encouraged to upgrade to v3.1 as this version introduces full support for iOS 9.

ThreatMetrix SDK v3.1 and newer

This version will use the new profiling hostname by default. Customers that have explicitly referenced the old profiling infrastructure either via our generic hostname (h.online-metrix.net) or via Enhanced Profiling via Hosted SSL (whereby ThreatMetrix hosts an SSL certificate on behalf of the customer) should remove these references from their code when implementing ThreatMetrix SDK
v3.1. This will cause the SDK to automatically reference the new profiling hostname.

Customers may also choose to obtain a new Enhanced Profiling SSL certificate that meets these stronger SSL cipher requirements. This should be done **before**deploying ThreatMetrix SDK in iOS 9 apps. Please contact our support department by clicking the Technical Support button above.

Why is the SDK unable to communicate with the ThreatMetrix Platform?

Each customer must be enabled in the ThreatMetrix Platform in order for the SDK to correctly communicate with our platform.

If you are seeing issues communicating with the ThreatMetrix platform, please contact ThreatMetrix Technical Support.

What do I do if I see an error in Xcode 10.1 and newer?

Xcode 10.1 and newer versions are printing the following error when using SDK:

[NetworkInfo] Signal strength query returned error: Error Domain=NSPOSIXErrorDomain Code=13 "Permission denied", descriptor: <CTServiceDescriptor 0x28051d700, domain=1, instance=1>

This error can be safely ignored https://forums.developer.apple.com/thread/111334. Customers

don't need to worry about this error and don't need to do anything, we can assure them that SDK is not accessing any hidden or forbidden functions. Xcode 10.1.

Use Questions

What type of scans does LexisNexis Risk Solutions perform on the ThreatMetrix Mobile SDK?

LexisNexis Risk Solutions performs various scans on the ThreatMetrix Mobile SDK, these scans are not disclosed to Customer's. LexisNexis Risk Solutions does not perform pen tests on the ThreatMetrix Mobile SDK. LexisNexis Risk Solutions does have Customer's who perform their own pen tests on the ThreatMetrix Mobile SDK. NOTE: Customer's must notify LexisNexis Risk Solutions when they plan to pen test the ThreatMetrix Mobile SDK.

What are the impacts of Google's requirement to target Android API 33?

Google has announced their new API level requirements for Advertising ID, which can be reviewed here: https://developer.android.com/about/versions/13/behavior-changes-13#ad-id

The new requirements state that Apps that use Google Play services advertising_id and target Android 13 (API level 33) and higher must declare the AD_ID normal permission in their app's manifest file, as follows:

```
<manifest ...>
    <!-- Required only if your app targets Android 13 or higher. -->
    <uses-permission
android:name="com.google.android.gms.permission.AD_ID"/>
    <application ...>
        ...
    </application>
</manifest>
```

If your app does not declare this permission when targeting Android 13 or higher, the advertising ID is automatically removed and replaced with a string of zeroes.

What are the impacts of Google's requirement to target Android API 30?

Google has announced their new API level requirements which can be reviewed here: https://developer.android.com/distribute/best-practices/develop/target-sdk

The new requirements state that starting in August 2021, new apps will need to target API level 30 (Android 11) or above and starting in November 2021, app updates will be required to target API level 30 or above. Existing apps that are not receiving updates are unaffected and can continue to be downloaded from the Play Store.

ThreatMetrix customers are highly recommended to update to the SDK >= 6.2 by November 2021 which can be found at SDK Downloads.
Staying on SDK 6.1 or 6.0 is not encouraged because of unexpected exceptions depending on the device state. If you would like to stay on SDK 6.1 or 6.0 versions, then two permissions will need to be added before November 2021.

- READ_PHONE_STATE
- ACCESS_FINE_LOCATION or ACCESS_COARSE_LOCATION

NOTE: Adding the above permissions might not be enough because the permissions can be revoked at any time, resulting in unexpected behavior, including crashes. SDK 5.4 or lower should not be used because it is not targeting API 30.

The Android M+ Permissions article provides more details regarding the permissions.

IMPORTANT:
As documented by Google, visibility into the inventory of installed apps is restricted when an application targets API 30. This means ThreatMetrix package scan feature can only access a limited list of applications (mainly ones installed by Google or phone manufacturers). To be able to use package scan as before, customers which fall into exception policies can include QUERY_ALL_PACKAGES permission in their application. Please see "Permitted uses of the 'query all packages' permission" and "Exceptions" sections in the above document to check if your application can use QUERY_ALL_PACKAGES permission.

Will users of Android Q be impacted?

If end users use Android Q, then please note the following:

- End users using Android Q might get a false positive "RandomisedID"
- Customers need to include **ACCESS_FINE_LOCATION**in order to use wifi positioning.
- device_imei information will not be available.
- device_id might change due to privacy restrictions in Android Q.

What ports and protocols does ThreatMetrix SDK use?

The following ports and protocols are used by ThreatMetrix SDK during profiling:

Port	Protocol	Notes
443	HTTPS / TCP	Used to transmit the profiling data from the device to the ThreatMetrix platform.
8080	HTTPS TCP	Used for Proxy Piercing. A connection is established from the client to the ThreatMetrix platform to expose the True IP address of the device.
53	DNS	Used for DNS IP detection. A connection is established from the device to their configured DNS server, which in turn queries a specialized DNS server in the ThreatMetrix platform to expose the DNS Server IP address used by the device. Theoretically, the DNS server in use by the device should be within a close proximity to the True IP address of the device.

Does ThreatMetrix SDK use WebView?

No. Starting with iOS SDK 5.1 and Android SDK 6.0, the ThreatMetrix SDK no longer uses WebView.

Does the ThreatMetrix SDK use the iOS Advertising Identifier?

No. ThreatMetrix SDK does not use the iOS Advertising Identifier (IDFA). The use of the IDFA was retired in SDK v2.0. The Apple sanctioned Vendor ID is now being used instead.

The discontinuation of the IDFA usage does not in any way impact the existing identifiers that have been generated and assigned to devices, nor does it weaken our device identification.

References:
iOS Documentation for IDFA.
iOS Documentation for Vendor ID.

Is ThreatMetrix SDK compliant with Google and Apple app store publishing guidelines?

Yes. ThreatMetrix SDK adheres to all required publishing, advertising and privacy guidelines for both Google and Apple. Under no circumstances will your application be blocked at submission time, or retrospectively removed from the respective marketplaces as a result of ThreatMetrix code. The guidelines for both vendors are listed below:

Google
Google Play Developer Program Policies: https://play.google.com/about/developer-content-policy.html
Developer Distribution Agreement: https://play.google.com/about/developer-distribution-agreement.html

Apple
App Review: https://developer.apple.com/app-store/review/
App Store Review Guidelines (Requires Apple Developer Login): https://developer.apple.com/appstore/resources/approval/guidelines.html

Product Guides

Introduction to Behavioral Biometrics

Overview

> **Note**
> Behavioral Biometrics requires an annual license.
> Please contact ThreatMetrix® Sales or Professional Services for more information.

Behavioral Biometrics provides the ability to collect signals based on how users utilize their mouse, keyboard, or touchscreen to interact with a web page or mobile app. By collecting and analyzing these signals, ThreatMetrix® is able to provide additional indicators for fraud, bot activity, social engineering, and remote access, and for the identification of returning users.

Behavioral Biometrics data is exposed both as individual attributes and as risk scores that can be utilized in policies and custom models. Below are examples of some of the data collected by Behavioral Biometrics. For a full list of attributes and scores derived from Behavioral Biometrics signals, please see: Behavioral Biometrics Attributes.

> **Note**
> Behavioral Biometrics was designed with privacy in mind.
> This means that NO password or PII is captured, thereby maintaining the privacy and security of all end users.

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Mouse Off-Page: Number of times mouse cursor moved off-page.

Access Method: How each input field was accessed (mouse, tab, shiftTab).

Timing: Timestamps and duration for all actions detected on input fields.

Remote Access: Rem detection is enhanced movement indicators.

of
ll

Shortcuts: Use of cut, copy, paste.

Auto: Autofill or autocompletion was used to populate an input field.

Errors: Number of times backspace was used to correct errors.

Risk Scores: Scores der Behavioral Biometrics ir different MOs.

Implementation

A standard ThreatMetrix® tags.js implementation is required for Behavioral Biometrics to operate on desktop and mobile browsers. Once enabled for your **org_id**, no additional JavaScript tags or

libraries are required; you will start seeing Behavioral Biometrics data as part of the Session Query API response and as additional fields in the Portal.

ThreatMetrix® also offers a Behavioral Biometrics module for mobile apps as of SDK 6.1. Please reach out to ThreatMetrix® Professional Services for enablement and implementation.